

$$\begin{aligned}
 Q3 \quad \hat{S}_n &= \frac{\hbar}{2} (\sin\theta \cos\phi \sigma_x + \sin\theta \sin\phi \sigma_y + \cos\theta \sigma_z) \\
 &= \frac{\hbar}{2} \left(\begin{pmatrix} 0 & \sin\theta \cos\phi \\ \sin\theta \cos\phi & 0 \end{pmatrix} + \begin{pmatrix} 0 & -i \sin\theta \sin\phi \\ i \sin\theta \sin\phi & 0 \end{pmatrix} + \begin{pmatrix} \cos\theta & 0 \\ 0 & -\cos\theta \end{pmatrix} \right) \\
 &= \frac{\hbar}{2} \begin{pmatrix} \cos\theta & \sin\theta \cos\phi - i \sin\theta \sin\phi \\ \sin\theta \cos\phi + i \sin\theta \sin\phi & -\cos\theta \end{pmatrix}
 \end{aligned}$$

$$\begin{aligned}
 \text{Let } \hat{S}_n |\psi\rangle &= \lambda |\psi\rangle \\
 \det(\hat{S}_n - \lambda I) &= 0 \\
 \begin{vmatrix} \frac{\hbar}{2} \cos\theta - \lambda & \frac{\hbar}{2} (\sin\theta \cos\phi - i \sin\theta \sin\phi) \\ \frac{\hbar}{2} (\sin\theta \cos\phi + i \sin\theta \sin\phi) & -\frac{\hbar}{2} \cos\theta - \lambda \end{vmatrix} &= 0 \\
 -\left(\frac{\hbar}{2} \cos\theta - \lambda\right) \left(\frac{\hbar}{2} \cos\theta + \lambda\right) - \frac{\hbar^2}{4} (\sin\theta \cos\phi - i \sin\theta \sin\phi) (\sin\theta \cos\phi + i \sin\theta \sin\phi) &= 0 \\
 -\frac{\hbar^2}{4} \cos^2\theta + \lambda^2 - \frac{\hbar^2}{4} (\sin^2\theta \cos^2\phi + \sin^2\theta \sin^2\phi) &= 0 \\
 -\frac{\hbar^2}{4} + \lambda^2 &= 0 \\
 \lambda &= \pm \frac{\hbar}{2}
 \end{aligned}$$

When $\lambda = \frac{\hbar}{2}$,

$$\hat{S}_n \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \frac{\hbar}{2} \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$$

$$\frac{\hbar}{2} \begin{pmatrix} \cos\theta - 1 & \sin\theta \cos\phi - i \sin\theta \sin\phi \\ \sin\theta \cos\phi + i \sin\theta \sin\phi & -\cos\theta - 1 \end{pmatrix} \begin{pmatrix} \alpha \\ \beta \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\Rightarrow (\cos\theta - 1)\alpha + (\sin\theta \cos\phi - i \sin\theta \sin\phi)\beta = 0$$

$$\frac{\alpha}{\beta} = \frac{\sin\theta \cos\phi - i \sin\theta \sin\phi}{1 - \cos\theta}$$

$$\sqrt{(1 - \cos\theta)^2 + (\sin\theta \cos\phi - i \sin\theta \sin\phi)^2}$$

$$= \sqrt{1 - 2\cos\theta + \cos^2\theta + \sin^2\theta \cos^2\phi - 2i \sin^2\theta \sin\phi \cos\phi - \sin^2\theta \sin^2\phi}$$

$$= \sqrt{4 \sin^4 \frac{\theta}{2} + \sin^2\theta e^{i2\phi}}$$

$$= \sqrt{4 \sin^4 \frac{\theta}{2} + 4 \sin^2 \frac{\theta}{2} \cos^2 \frac{\theta}{2} e^{-i2\phi}}$$

$$= 2 \sin \frac{\theta}{2} \sqrt{\sin^2 \frac{\theta}{2} + \cos^2 \frac{\theta}{2} e^{-i2\phi}}$$

$$= 2 \sin \frac{\theta}{2} \sqrt{1 + \cos^2 \frac{\theta}{2} (e^{-i2\phi} - 1)}$$

$$= 2 \sin \frac{\theta}{2} \sqrt{1 - \cos^2 \frac{\theta}{2} (2 \sin^2 \phi + 2 \sin\phi \cos\phi i)}$$

$$= 2 \sin \frac{\theta}{2} \sqrt{1 - 2 \sin^2 \phi \cos^2 \frac{\theta}{2} e^{i\phi}}$$

$$\alpha = \frac{e^{i\phi} \cos \frac{\theta}{2}}{\sqrt{1 - 2 e^{i\phi} \sin^2 \phi \cos^2 \frac{\theta}{2}}}, \quad \beta = \frac{\sin \frac{\theta}{2}}{\sqrt{1 - 2 e^{i\phi} \sin^2 \phi \cos^2 \frac{\theta}{2}}}$$

$$\therefore |\psi_+\rangle = \frac{e^{i\phi} \cos \frac{\theta}{2}}{\sqrt{1 - 2 e^{i\phi} \sin^2 \phi \cos^2 \frac{\theta}{2}}} |+\rangle + \frac{\sin \frac{\theta}{2}}{\sqrt{1 - 2 e^{i\phi} \sin^2 \phi \cos^2 \frac{\theta}{2}}} |-\rangle$$